

Storytelling Script

Bitcoin Volatility Prediction — Technical-Review Deck (17 slides, ~18 minutes)

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How to Use This Script

This script is written for the restructured 17-slide deck (`capstone-btc-volatility-canva.pptx`) — the one that follows a technical-review flow: objective, data, EDA, feature selection, models, comparison, limitations, implications, conclusion.

It's written as a *story*, not a slide-by-slide recitation. There's a real narrative spine running through it: you go in with a hypothesis, the data pushes back on part of it, you find the one clue that actually matters, you test three different approaches against that constraint, and you end with an honest verdict about what the result is actually good for. Say each slide's beats in your own words — the through-lines between slides (the “but,” the “here's the twist,” the callback) are the part worth keeping close to as written, since those are what make it feel like one story instead of seventeen facts.

[pause] marks a spot to let something land — a number, a reveal, a turn. These are the moments the story lives or dies on; don't rush them.

Total target time: ~18 minutes, comfortable inside a 20-minute slot.

Slide 1 — Title

Target: 0:25 (cumulative 0:25)

Good [morning/afternoon]. I'm Mohammed Hejairi, and this is Bitcoin Volatility Prediction.

[pause]

Here's the question this whole project is built around: not “will Bitcoin go up or down” — nobody can answer that reliably — but “how *rough* is the ride going to be.” That turns out to be a very different, and much more answerable, question. This is the story of how far you can actually push that answer — and, just as importantly, where it stops.

Slide 2 — Objective: 7-Day Volatility Forecasting

Target: 1:00 (cumulative 1:25)

Let me tell you what I set out to do, and — more unusually — what I deliberately did *not* set out to do.

The goal: forecast Bitcoin’s 7-day realized volatility from hourly price data. Not the price. The *turbulence*.

[pause]

And here’s the part I want to say plainly, right at the start, because it shapes everything that follows: the objective here was never to walk in here and claim some incredible crystal-ball accuracy. It was to show my work honestly — to let the exploratory analysis and the feature engineering set realistic expectations *before* a single model got built, so that whatever number comes out at the end has a real, defensible story behind it instead of just being a number.

That gives the whole project a four-act structure: what does the data actually tell us — what signal can we actually extract from it — can we turn that into a working forecast — and, honestly, what would it take to do better. That’s the story I’m about to walk you through.

Slide 3 — Data Overview & Feature Dictionary

Target: 0:50 (cumulative 2:15)

First, the raw material. Over 73,000 hours of Bitcoin trading data — that’s every hour, going back to 2017. Price, volume, and a set of features I engineered specifically to measure volatility at different time horizons: a day, a week, a month.

And the target — the thing we’re actually trying to predict — is volatility measured 7 days, 168 hours, *ahead* of everything we know. Not “what’s volatility right now.” What will it be a full week from this exact moment. That gap is the whole difficulty of this problem, and it’s going to matter again later.

Slide 4 — Dataset & Data Cleaning

Target: 0:50 (cumulative 3:05)

Before any of that data was usable, it needed cleaning — the unglamorous part nobody puts in the highlight reel, but the part that quietly decides whether everything downstream is trustworthy. Gaps in the exchange feed, zero-volume candles that would break the math, timestamp inconsistencies — all of it had to be caught and handled *before* it could quietly poison a result three steps later.

[pause]

And once it was clean, the target’s own shape told me something immediately: look at that distribution. Long calm stretches, piled up on the left — and then a long thin tail stretching out to

the right. That's not noise. That's volatility clustering, and it's the first real clue in this whole investigation.

Slide 5 — Feature Engineering

Target: 1:00 (cumulative 4:05)

So I built four tools to try to capture that pattern. Volatility measured over a day, a week, and a month — layering short and long memory together. A range-based measure, ATR, rescaled as a percentage of price instead of raw dollars. A smoothed exponential estimate. And realized skewness, which captures the *shape* of returns, not just their size.

[pause]

I want to flag one of these honestly, because it's a real story on its own: that ATR rescaling wasn't a nice-to-have. The raw dollar-value version was off by a factor of roughly 45,000 against every other regressor — a genuine bug — and fixing it turned out to be the single biggest improvement in the entire project. Sometimes the breakthrough isn't a fancier model. It's just noticing the units were wrong.

Slide 6 — EDA: Temporal Features

Target: 1:10 (cumulative 5:15)

Now the investigation really starts, and it splits into two threads — one that goes nowhere, and one that goes everywhere.

The dead end first: does *when* something happens — the hour of day, the day of the week — tell you anything about how severe it'll be? I ran the numbers. Chi-square statistic of 2.68, 46 degrees of freedom, p-value of **[pause]** 1.000000. That is about as close to “absolutely no signal” as a statistical test can say. The calendar doesn't matter here.

But the other thread — volatility's own recent history — decays slowly across many lags instead of dropping off a cliff. That's real, exploitable memory. That single distinction is why every serious regressor in this project measures volatility itself, and not a single one of them cares what hour it is.

Slide 7 — EDA: Range & Dispersion Features

Target: 0:55 (cumulative 6:10)

Next lead: range-based measures — how far price traveled within each candle, not just where it closed. Two of them, ATR and Bollinger width, both built from intraday highs and lows.

[pause]

Bollinger width correlates with the target at plus-0.588 — by far the strongest number in this whole analysis. ATR sits at plus-0.142. Promising. But — and this is the catch — they're both measuring almost the same underlying thing. This isn't two independent tips. It's one good tip, told twice.

Slide 8 — EDA: Volatility Clustering

Target: 0:50 (cumulative 7:00)

So does that clustering pattern actually let you draw a clean line between “calm” and “turbulent”? Look at the same distribution again, now split into three: blue, calm; gold, moderate; red, turbulent.

[pause]

They blend into each other. There's no sharp edge — just a gradient. Clustering is real, and it's exploitable — but it sets a *ceiling* on how well any of this can work. It doesn't hand you a guarantee.

Slide 9 — Feature Selection: Correlation & Effect Sizes

Target: 1:05 (cumulative 8:05)

This is the turning point of the whole story — the moment where the investigation stops chasing leads and delivers a verdict.

Four features earned a place in the final model: the three time-horizon volatility measures, plus the rescaled ATR, plus the smoothed and skew-based additions. Everything else — RSI, MACD, raw price levels — got dropped. Weak, redundant, or flat-out spurious.

[pause]

And here's the critical finding, the one that explains everything that comes after it: even the *strongest* correlation we found — that 0.588 for Bollinger width — turns out to be redundant with ATR, not new information on top of it. Which means the ceiling on accuracy isn't a limitation of any model I'm about to show you. It was set right here, by the data itself, before I'd trained a single thing.

Slide 10 — Feature Relationships: Correlation Analysis

Target: 0:45 (cumulative 8:50)

Here's that same finding as one picture. Two features with real signal — `bb_width`, `atr_14` — and then a long tail of everything else, sitting at essentially zero. That gap you're looking at is the performance ceiling. Models cannot manufacture information that was never in the data to begin with.

Slide 11 — Model 1: GARCH

Target: 1:00 (cumulative 9:50)

With that ceiling now understood, it's time to actually go test it — three different approaches, three different bets on how to extract what little signal exists.

First up: GARCH, tuned into its GJR variant with Student-t errors — the textbook, industry-standard volatility model, used everywhere in finance.

[pause]

R-squared: **negative 5.02**. And on genuinely unseen live data, it collapses even further, to negative 110. Not a bug — GARCH's forecast structurally mean-reverts toward a long-run average well before a full week is up. It's the right tool for a different job, and this is the first real casualty of the story.

Slide 12 — Model 2: HAR-RV-X

Target: 1:05 (cumulative 10:55)

Second attempt: HAR-RV-X — a regression across those three volatility horizons, Ridge-regularized, with a log transform on the target.

[pause]

R-squared: **0.242**. And — this is the number that actually matters most — it holds up on live data too, at 0.178. This is the winner. Not because it's the most sophisticated model in the lineup. Because it's built directly around the one real signal the investigation actually found.

Slide 13 — Model 3: EWMA

Target: 0:50 (cumulative 11:45)

Third: EWMA — the RiskMetrics-style exponential smoothing every risk desk on Wall Street already trusts. One parameter, backtested to 0.995.

[pause]

R-squared: 0.059. A real, honest runner-up — simple, transparent, no hidden layers — just not as sharp as a model built around three time horizons at once instead of one.

Slide 14 — Model Comparison

Target: 1:05 (cumulative 12:50)

Put all three side by side against the baselines, and the story resolves itself. HAR-RV-X wins on every single metric, beating simple naive persistence by 17.1% lower error. EWMA also clears that bar. GARCH and Holt-Winters do not.

[pause]

And the detail I want to land on: this ranking isn't just true on the historical test set I built it on. It holds on live data the model had never touched during training. That's the difference between a result that looks good and a result that's actually real.

Slide 15 — Limitations

Target: 1:10 (cumulative 14:00)

Now, the honest part of the story — the part a lot of presentations skip.

Three categories of limitation. Data: no fundamentals, no on-chain signals, no other coins — just price and volume history. Modeling: twenty-two separate robustness checks were run, and most of them did not beat HAR-RV-X — that's not a footnote, that's the actual record of what was tried. Statistical: the feature correlations were weak to begin with, so no amount of model cleverness was ever going to lift this past a hard ceiling.

[pause]

The honest takeaway: real improvement here needs genuinely *new* information — news sentiment, on-chain data — not more tuning of what's already on the table.

Slide 16 — Operational Implications

Target: 1:05 (cumulative 15:05)

So given all of that — is this actually useful? Yes, but specifically, not generally.

Appropriate: sizing a position so a bigger expected swing means a smaller bet. Feeding one data-driven number into options pricing, to compare against what the market is charging. A continuous input for a weekly risk report.

Inappropriate: trading on this number alone — it explains about a quarter of what actually happens. Real-time, automated execution — it was never built or tested for that. And definitely not treating it as a guarantee in any legal or compliance context.

[pause]

One line sums up the whole slide: this is one input. Not the final answer.

Slide 17 — Conclusion

Target: 0:45 (cumulative 15:50)

So here's the moral of the story. Every time a more flexible, more powerful model — XGBoost, an LSTM, ARIMA — went up against HAR-RV-X's three purpose-built regressors, it lost. Not because those models were weaker in general. Because the *information* was the bottleneck the whole time, not the architecture trying to use it.

[pause]

Purpose-built regressors. Not more complexity. That's the finding — and it's a more useful one than a flashier model would have given me. Thank you — happy to take questions.

Pacing Checkpoints

Checkpoint	Slide	Target elapsed time
Premise set	End of Slide 2	~1:25
Data introduced, groundwork laid	End of Slide 5	~4:05
Investigation complete (EDA)	End of Slide 8	~7:00
Turning point — the ceiling found	End of Slide 10	~8:50
All three models tested	End of Slide 13	~11:45
Verdict delivered	End of Slide 14	~12:50
Honesty + real-world payoff	End of Slide 16	~15:05

Checkpoint	Slide	Target elapsed time
Finished	Slide 17	~15:50

Note: this leaves roughly 4-5 minutes of slack inside a 20-minute slot — plenty of room for a slower pace on the day, or for pausing to let a number sit before moving on.